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Evaluating the effectiveness of advanced platelet-rich fibrin, photobiomodulation, pentoxifylline, and Alveogyl in the treatment of alveolar osteitis: a randomized controlled clinical trial

Alper Yuzbasioglu[1] and Cennet Neslihan Eroglu[1*]

Abstract

Background Alveolar osteitis is a type of small-scale osteomyelitis of the alveolar bone that occurs after tooth extraction, the etiology of which remains unknown, and alternative methods are being investigated for its treatment. The aim of this study was to compare the effectiveness of advanced platelet-rich fibrin (A-PRF), photobiomodulation (PBM), and Alveogyl (butamben, idoform, eugenol), which have shown success in the treatment of alveolar osteitis, with that of pentoxifylline (PTX) to determine whether PTX could be an alternative treatment for alveolar osteitis.

Methods This study included 80 healthy volunteers diagnosed with alveolar osteitis in the extraction sockets of their mandibular first, second, and third molars. The patients were divided into four groups, with 20 patients in each group: A-PRF, PBM, PTX, and Alveogyl. After physiological saline irrigation, the patients were treated according to their respective group. The patients were followed up on the 2nd, 4th, 7th, and 14th days. Primary outcomes included pain assessment and the evaluation of soft tissue healing. Pain was assessed via the visual analog scale (VAS), soft tissue healing was evaluated via the Landry Healing Index (LHI), and granulation tissue was measured. Age and sex were used as study variables. The data were analyzed via ANOVA and post hoc tests to compare the treatment groups.

Results There was no statistically significant difference in the mean VAS score between the groups ($p > 0.05$). However, the degree of reduction in the VAS score increased in the following order: A-PRF, PBM, PTX, and Alveogyl. In the LHI data, there was a statistically significant difference in the mean scores between the groups, with Alveogyl being favored preoperatively and PBM being favored at the 2nd week ($p < 0.05$). Statistically significant differences were observed in granulation tissue measurements on the preoperative day, as well as on Days 4 and 7, in favor of Alveogyl on Day 4 and PBM on Day 7 ($p < 0.05$). The greatest increase in granulation tissue scores was observed in the A-PRF group, whereas the greatest change in LHI scores was observed in the PBM group. PTX did not produce a statistically significant difference in soft tissue healing, although there were periods when it showed results similar to those of Alveogyl and PBM. There were significant differences between the VAS score, granulation tissue score and LHI score in terms of age and sex within the time periods followed.

*Correspondence:
Cennet Neslihan Eroglu
neslihanakca2003@yahoo.com
Full list of author information is available at the end of the article

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Conclusion The findings of this study indicate that although PTX has similar efficacy in terms of wound healing and analgesic properties to the methods used, it does not offer significant advantages. Compared with other methods, A-PRF and PBM have provided better results in the treatment of alveolar osteitis, particularly regarding pain and soft tissue healing.

Trial registration The study was retrospectively registered in the clinical trial registry with the number TCTR20231014003 on 14.10.2023.

Keywords Alveolar osteitis, Dry socket, Alveogyl, Photobiomodulation, A-PRF, Pentoxifylline

Introduction

Alveolar osteitis, or dry socket, is defined as the presence of postoperative pain that intensifies between 1 and 3 days after extraction once other potential causes of pain have been ruled out. It is characterized by partial or complete loss of the blood clot in the extraction socket and is sometimes accompanied by halitosis [1]. Although the exact cause of alveolar osteitis is not fully understood, various factors, such as trauma during extraction, bacterial infection, and biochemical agents, along with systemic predisposing factors, are believed to contribute to its etiology [2]. It has been proposed that trauma during tooth extraction induces local inflammation at the socket, triggers the release of tissue activators, and increases plasmin levels, which can lead to dissolution of the blood clot in the socket [3]. Furthermore, surgical trauma is thought to delay wound healing due to traumatic thrombosis of blood vessels and increased susceptibility to infection due to reduced tissue resistance [4].

In addition to avoiding risk factors, various treatments, such as Alveogyl agents, antifibrinolytic agents, antibiotics, analgesics, antiseptic agents, platelet products, photobiomodulation (PBM), photodynamic therapy, or combinations thereof, are used to prevent the development of alveolar osteitis [5].

Alveogyl is a product containing eugenol, sodium lauryl sulfate, calcium carbonate, and auxiliary components; its previously marketed form, known as Alvogyl, also contains butamben and iodoform [6]. This compound is most commonly preferred by clinicians in the conventional treatment of alveolar osteitis [7].

Photobiomodulation facilitates increased oxygen delivery to the area and enhances the passage of immune cells into tissues by inducing vasodilation and increasing local blood flow. It promotes relaxation and relief in smooth muscles of the endothelium, thereby facilitating rapid tissue healing [8, 9]. There are few studies in the literature on the application of PBM in the treatment of alveolar osteitis. Previous studies have shown that PBM application is an effective method for pain relief and promoting socket healing in patients with alveolar osteitis [10–12]. PBM has been reported to be significantly more effective in alleviating alveolar osteitis pain compared to treatments such as Salicept, zinc oxide eugenol, ozone therapy, and photodynamic therapy [13].

Autologous blood products support healing by promoting angiogenesis and exerting immunomodulatory effects [14]. Compared with platelet-rich fibrin (PRF), another method that has gained popularity recently, advanced platelet-rich fibrin (A-PRF), has demonstrated superior performance in increasing the number of inflammatory cells and releasing growth factors. A-PRF is an advantageous platelet product for tissue regeneration [15]. In the literature, only one study has utilized A-PRF for the treatment of alveolar osteitis, and A-PRF has been found to be effective in promoting wound healing and reducing pain [16].

Pentoxifylline (PTX) is a derivative of methylxanthine (1-(5-oxohexyl)-3,7-dimethylxanthine) used to treat peripheral vascular diseases. PTX acts as a vasoactive agent, enhancing peripheral blood flow and tissue oxygenation by increasing erythrocyte elasticity [17]. It also promotes collagenase activation. Because of these effects, PTX is utilized not only for treating peripheral arterial diseases but also for managing radiation-induced tissue damage, promoting bone continuity, accelerating soft tissue healing, and reducing wound dehiscence [18, 19].

While studies continue to investigate the prevention of alveolar osteitis, which can occur unpredictably despite a high clinical probability or minimal predisposing factors, there remains a necessity for research into alternative treatments aimed at alleviating pain and reducing symptomatic days for affected patients.

The limited number of studies involving A-PRF for the treatment of alveolar osteitis creates a gap in the literature that must be addressed to evaluate whether PTX could be a viable option for secondary wound healing in patients with alveolar osteitis. This study sought to explore whether PTX could serve as an alternative therapeutic approach to A-PRF, PBM, and Alveogyl in the management of alveolar osteitis.

Null Hypothesis (H_0): There is no significant difference in the effectiveness of A-PRF, PBM, Alveogyl, or PTX in the treatment of alveolar osteitis in terms of pain reduction, soft tissue healing (LHI), or granulation tissue formation.

Materials and methods

The study enrolled volunteers aged between 18 and 65 years, without systemic diseases, who reported pain following tooth extraction. Participants were recruited from the Department of Oral and Maxillofacial Surgery, Faculty of Dentistry, Akdeniz University.

The procedures were explained to all participants, and written informed consent was obtained from all of them before they participated in the study. The study was approved by the Clinical Research Ethics Committee of the Faculty of Medicine of the University (KAEK 21.09.2022/253–26.04.2023/155) and the Turkish Medicines and Medical Devices Agency (30.05.2023/1120566). All procedures were performed in accordance with the principles outlined in the 1964 Helsinki Declaration and its later amendments. The study was retrospectively registered in the clinical trial registry with the number TCTR20231014003 on 14.10.2023 and adheres to the recommendations of the CONSORT 2010 statement for reporting randomized trials.

The inclusion criteria for the study were as follows: cases of alveolar osteitis meeting the standardization criteria described by Blum [20] and developing 1–3 days after tooth extraction; patients who had not received treatment for alveolar osteitis; patients without systemic disease; patients without active infection around the extraction socket; and patients whose extraction sockets of mandibular first and second molars or mandibular wisdom teeth healed via secondary intention without indications for suturing.

The exclusion criteria for the study included patients using oral contraceptives, pregnant or breastfeeding women, smokers or alcohol consumers, individuals unable to provide informed consent, and patients with incomplete examination or follow-up records.

Based on a previous study [16], a power analysis was conducted with a Type I error rate of 5% and 95% power using a repeated measures design. The sample size estimation for the study was performed with G*Power 3.1.9.2 software [21]. For block randomization, blocks were formed based on individuals under 40 years of age and those 40 and over, as well as by sex. The initial random assignments were made to Groups 1 and 3, followed by subsequent assignments to Groups 2 and 4.

Study groups

After completing the preoperative evaluations and obtaining records, the patients were divided into four groups, each comprising 20 patients. The following treatment protocols were applied accordingly: Group 1 received A-PRF, Group 2 received PBM, Group 3 received PTX, and Group 4 received Alveogyl (Fig. 1).

In Group 1, after irrigation with 10 ml of saline solution, 10 ml of the patient's blood was collected into a plastic tube without anticoagulant. The blood was centrifuged at 1500 rpm for 14 min (Hettich EBA 20 Zentrifuger, D-78532, 2006, Tuttlingen, Germany) to obtain A-PRF, which was then placed into the extraction socket [15]. The A-PRF + clot, located at the top of the 4 ml layer in the centrifuge tube, was first separated from the red thrombus and then positioned into the extraction socket, as described in the study by Yüce and Komerik [16]. Topical anesthesia was applied, and the area was sutured.

In Group 2, after irrigation with 10 ml of saline solution, photobiomodulation (PBM) was performed intraorally via an 810 nm wavelength GaAlAs diode laser (Picasso Dental Diode Laser, AMD Lasers, USA) with a deep tissue tip. The laser was adjusted to a spot size of 300 nm diameter in non-contact continuous mode for 20 s at a fluence of 20–25 J/cm^2 and 1.0 W/cm^2 on the occlusal and buccal areas [22]. PBM sessions were conducted on postoperative Days 0, 2, 4, and 7.

In Group 3, after irrigation with 10 ml of saline solution, PTX (Trentilin 400 mg Retard tablets, Santa Farma Drug Ind. Istanbul, Turkiye) was administered orally at a dose of 400 mg tablet twice daily for one week.

In the Group 4 control group, irrigation with 10 ml of saline solution was followed by the application of Alveogyl (manufactured by Septodont India Pvt. Limited. Content: iodoform = 15.8 gm, eugenol = 13.7 gm, and butamben = 25.7 gm). The amount of Alveogyl used was approximately the size of the lentil, which was collected in a single attempt via sterile forceps.

Throughout the postoperative period, all patients were prescribed 500 mg of paracetamol four times daily. Follow-up visits were scheduled for all patients on Days 2, 4, 7, and 14.

Blinding

The clinician performing the measurements and assessments and the statistician performing the analyses were blinded.

Study variables

Pain assessment

After the applications, patients' pain levels on Days 0, 2, 4, and 7 were assessed via the Visual Analog Scale (VAS). Patients were asked to rate their pain on a scale from 0 to 10, where 0 indicated no pain and 10 indicated unbearable pain [23].

Assessment of wound healing

In the weeks following the operation, wound healing was evaluated via the Landry Healing Index (LHI) scale [24], which ranges from 0 (indicating very poor

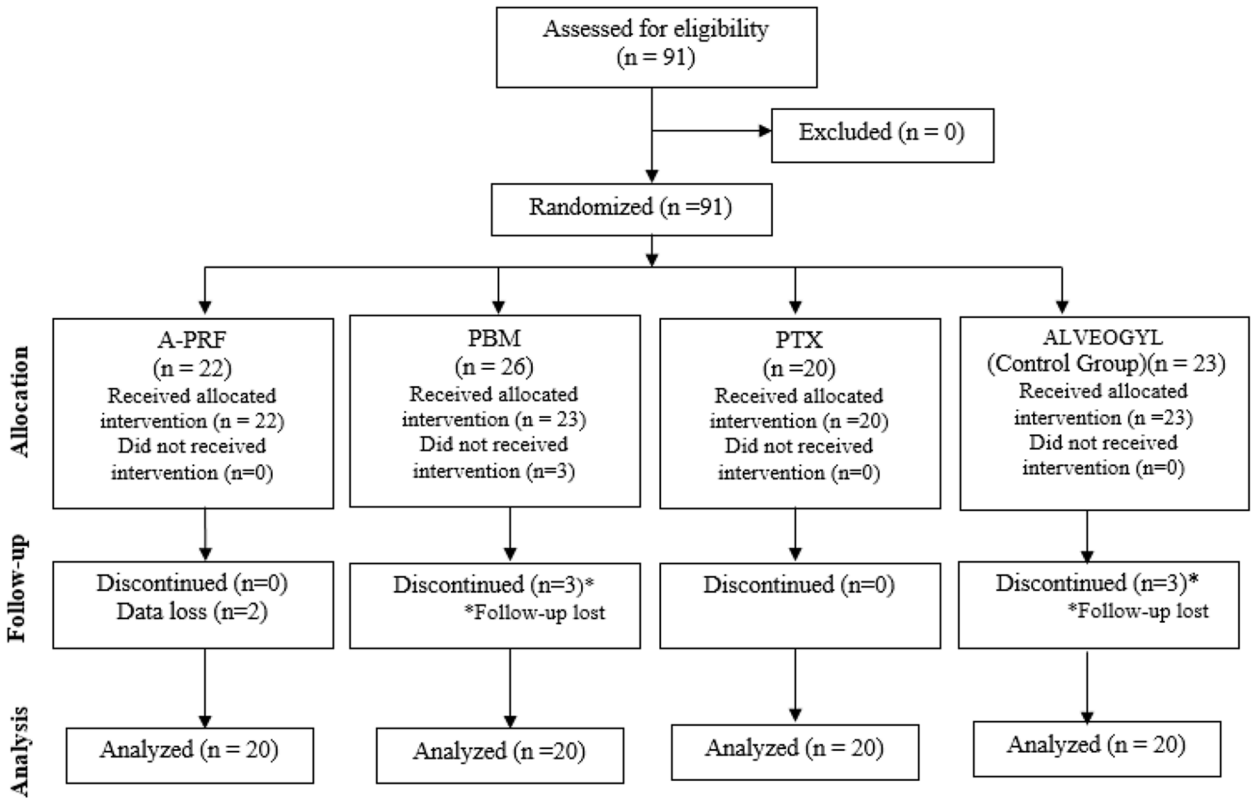


Fig. 1 Flow diagram of the single-center randomized parallel-group study

healing) to 4 (indicating excellent healing). Scores were interpreted as follows: 0 for very poor, 1 for poor, 2 for fair, 3 for good, and 4 for excellent healing.

Granulation tissue measurements were conducted on Days 0, 2, 4, and 7 postoperation, involving assessment of the tissues lining the walls of the alveolar socket [25]. Healing was categorized based on the extent of tissue formation: single-wall healing scored as 1, two-wall healing scored as 2, three-wall healing scored as 3, four-wall healing scored as 4, and the absence of healing scored as 0.

Statistical analysis

After the homogeneity of the data was assessed via the Kolmogorov–Smirnov test, the data were analyzed using frequency analysis, descriptive statistics, independent samples t tests, one-way analysis of variance (ANOVA), dependent samples t tests, and correlation analysis. Tukey-B (post hoc) multiple comparisons were conducted to identify groups that produced the difference. A p value < 0.05 was considered significant. IBM SPSS Statistics for Windows, Version 26.0. (IBM Corp., Armonk, NY, USA) was used for data analysis.

Results

In the present study, 91 patients were initially included, but 11 patients were subsequently excluded because of incomplete records and missed follow-up appointments. The study included a total of 80 patients. Details of the variables and percentages of patients are presented in Table 1. The mean ages ± standard deviations and sex percentages of the groups were as follows: A-PRF: 43.25 ± 15.50 (65% female to 35% male), PBM: 35 ± 11.46 (60% female to -40% male), PTX: 29.15 ± 8.44

Table 1 Variable distribution in the study

	Variables	n	%
Sex	Female	50	62,5
	Male	30	37,5
Age distribution	Between 18–27 years	27	33,8
	Between 28–35 years	15	18,8
	Between 36–44 years	20	25,0
	45 years and older	18	22,5
Tooth distribution	1. and 2. mandibular molars	43	53.75
	Mandibular wisdom teeth	37	46,25

(60% female to 40% male), and Alveogyl: 37.45 ± 13.04 (65% female to 35% male).

Throughout the study, no healing issues or PTX-related side effects were encountered.

Statistical analysis did not reveal a significant difference in the mean VAS score between the treatment groups. The extent of the decrease in the VAS score increased as follows: A-PRF, PBM, PTX, and Alveogyl (Table 2).

A significant difference in the mean VAS score was observed between the sexes, with higher preoperative and 4th scores among female patients. However, the VAS scores on the 2nd and 7th days were comparable between the sexes (Table 3).

When analyzed according to age groups, VAS scores were similar across the follow-up periods ($p > 0.05$). The evaluation of granulation tissue scores by age group revealed no significant differences in the preoperative, 4th-day, or 7th-day scores. However, analysis of 2nd-day granulation tissue scores revealed that individuals aged 45 years and older had higher average scores than those aged 18–27 years, 28–35 years, and 36–44 years ($p = 0.031$).

Similarly, while there were no differences between age groups in terms of LHI scores at the preoperative and 1st week assessments ($p > 0.05$), the average LHI score at the 2nd week for individuals aged 45 years and

Table 2 Comparison of mean VAS between groups

	Groups	n	$\bar{x}$	sd	F	sd	*p*
VAS Mean of Day 0	ALVEOGYL	20	7,05	1,638	1,462	3,76	0,232
	A-PRF	20	8,05	1,638			
	PBM	20	7,40	1,789			
	PTX	20	7,70	1,174			
VAS Mean of Day 2	ALVEOGYL	20	5,15	2,084	1,106	3,76	0,352
	A-PRF	20	5,40	2,280			
	PBM	20	4,75	1,860			
	PTX	20	5,85	1,531			
VAS Mean of Day 4	ALVEOGYL	20	3,60	1,818	1,215	3,76	0,310
	A-PRF	20	3,25	2,074			
	PBM	20	3,00	2,026			
	PTX	20	4,05	1,395			
VAS Mean of Day 7	ALVEOGYL	20	1,50	1,051	0,704	3,76	0,552
	A-PRF	20	1,50	1,051			
	PBM	20	1,35	0,988			
	PTX	20	1,85	1,387			

Table 3 The assessment of the Mean VAS, granulation tissue, and LHI scores was conducted over time, considering age groups and sex

	Female	Male	*p*	Age between 18–27	Age between 28–35	Age between 36–44	45 years and older	*p*
VAS Day 0	7,92 ± 1,426	6,93 ± 1,680	**0,006***	7,70 ± 0,993	7,27 ± 1,944	7,85 ± 1,631	7,22 ± 1,957	0,539
VAS Day 2	5,56 ± 2,002	4,83 ± 1,840	0,109	5,63 ± 1,471	5,20 ± 2,305	5,30 ± 1,922	4,83 ± 2,383	0,620
VAS Day 4	3,76 ± 1,890	3,00 ± 1,722	**0,049***	3,63 ± 1,548	3,20 ± 1,897	3,50 ± 1,821	3,44 ± 2,357	0,917
VAS Day 7	1,64 ± 1,191	1,40 ± 1,003	0,358	1,74 ± 1,289	1,40 ± 1,121	1,50 ± 1,051	1,44 ± 0,984	0,750
Granulation Day 0	0,38 ± 0,635	0,40 ± 0,563	0,887	0,37 ± 0,629	0,33 ± 0,488	0,40 ± 0,681	0,44 ± 0,616	0,960
Granulation Day 2	0,84 ± 0,618	0,63 ± 0,669	0,164	0,48 ± 0,643	0,80 ± 0,561	0,90 ± 0,641	1,00 ± 0,594	**0,031***
Granulation Day 4	1,22 ± 0,764	0,97 ± 0,615	0,127	1,00 ± 0,784	1,13 ± 0,640	1,20 ± 0,768	1,22 ± 0,647	0,721
Granulation Day 7	1,58 ± 0,883	1,53 ± 0,776	0,812	1,41 ± 0,694	1,73 ± 0,799	1,70 ± 0,923	1,50 ± 0,985	0,544
LHI Preop	0,70 ± 0,463	0,70 ± 0,535	0,998	0,74 ± 0,447	0,73 ± 0,458	0,55 ± 0,510	0,78 ± 0,548	0,463
LHI Week 1	1,68 ± 0,587	1,60 ± 0,563	0,551	1,63 ± 0,565	1,67 ± 0,617	1,60 ± 0,503	1,72 ± 0,669	0,926
LHI Week 2	2,48 ± 0,677	2,33 ± 0,606	0,333	2,30 ± 0,609	2,27 ± 0,594	2,40 ± 0,598	2,78 ± 0,732	**0,049***

*Statistically significant at $p<0.05$

older was greater than that for those aged 18–27 years, 28–35 years, and 36–44 years ($p=0.049$) (Table 3).

In the granulation tissue analyses conducted across patient groups, significant differences were observed on the preoperative, 2nd, 4th, and 7th days (Fig. 2). The mean granulation tissue scores were significantly greater in the Alveogyl group than in the A-PRF, PBM, and PTX groups at the preoperative time-point ($p=0.011$), on Day 4 ($p=0.019$), and on Day 7 ($p=0.005$). On Day 2, however, the PBM group presented significantly higher mean granulation tissue scores than the other groups did ($p=0.002$).

In the analysis of the LHI scores among the groups, statistically significant differences were found in favor of the Alveogyl group in the preoperative period ($p=0.001$) and in favor of the PBM group at the 2nd week ($p=0.000$). However, no significant difference was observed in the 1st week scores ($p=0.558$) (Fig. 3).

Discussion

Alveolar osteitis is characterized as minor, superficial osteomyelitis of the alveolar bone and influenced by a range of local and systemic factors [26]. The aim of this study was to explore the potential of the inclusion of PTX as a treatment option and compare its effectiveness

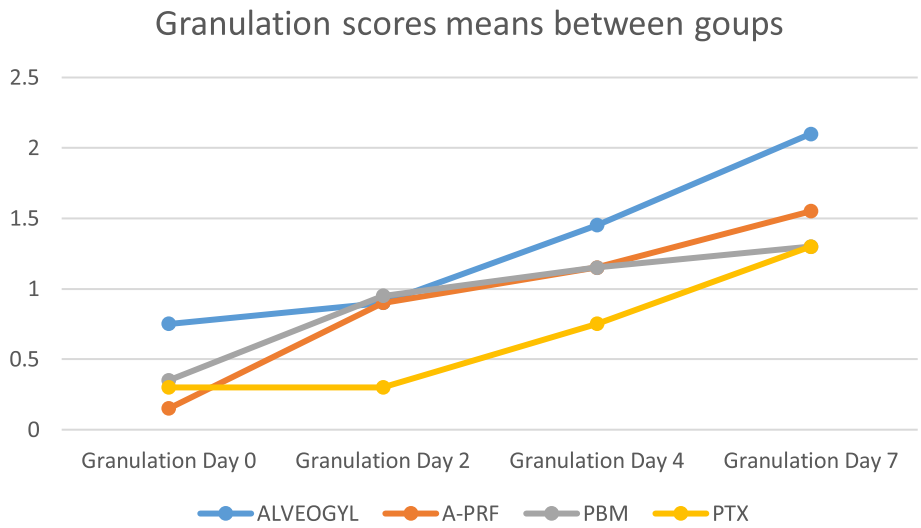


Fig. 2 Changes in the mean increase in granulation tissue scores according to the groups

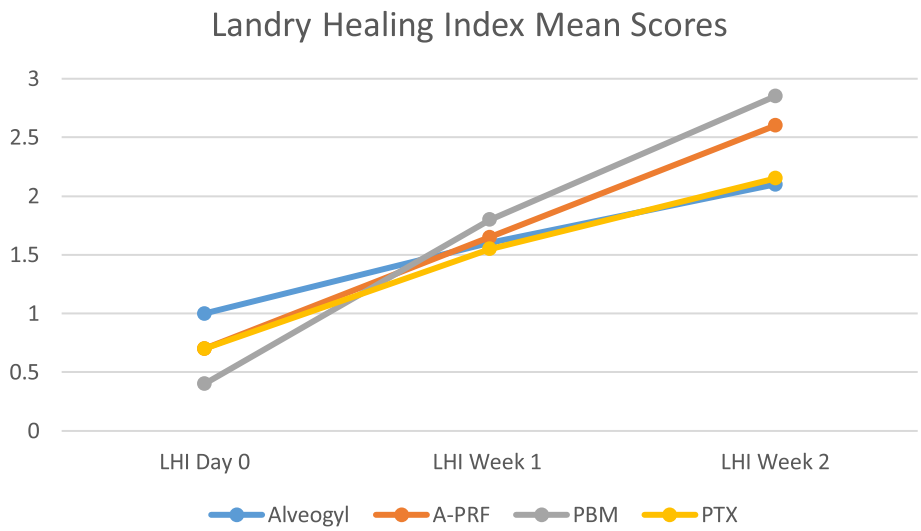


Fig. 3 Changes in the LHI score means according to group

with that of other established methods. According to the findings of this study, all the treatment groups exhibited comparable outcomes in terms of pain management. However, A-PRF and PBM were identified as the most effective methods for promoting soft tissue healing. The null hypothesis regarding pain was confirmed, but it was not validated in terms of soft tissue healing.

All the methods used in this study were successful in alleviating pain in patients with alveolar osteitis. Previous studies have shown Alveogyl, PBM, and A-PRF to be effective in terms of analgesic properties in cases of alveolar osteitis, which is consistent with the pain-relieving effectiveness results of this study. Although PTX is novel in this context and not primarily recognized for its analgesic effects, it effectively manages pain in patients with alveolar osteitis, possibly owing to its role in accelerating soft tissue healing [26]. Previous findings suggest that PTX shortens the inflammatory phase in the early stages of wound healing, leading to faster development of healing phases, although its effect may slow down in the later stages of wound healing [26]. In studies, PTX has been reported to be safe and associated with minimal side effects even with long-term use [27, 28]; consequently, it was administered for a limited duration in the present study to mitigate potential side effects associated with prolonged use. Patients tolerated PTX well throughout the treatment period. In this study, while PTX did not provide significant advantages, it may be considered a supportive medication in future clinical trials alongside existing treatment protocols. In general, it reached similar clinical efficacy levels as Alveogyl at the end of the 2nd week in the LHI evaluation and PBM at the end of the 7th day in the granulation tissue evaluation. Therefore, the difference that will occur in combined applications with other methods is likely to be at a level that will be reflected in statistics.

In studies on the treatment of alveolar osteitis, lower pain scores were observed with the application of PRF than with Alveogyl and zinc oxide eugenol (ZnOE) [29, 30]. PRF has also been found to be superior to Alveogyl in soft tissue healing [30]. In the only study in the literature conducted with A-PRF for treating alveolar osteitis, A-PRF was reported to significantly reduce pain and promote healing in both soft and bone tissues [16]. PRF products obtained at lower centrifugation speeds, such as A-PRF, arguably release higher levels of growth factors, support type-I collagen production, and thereby make significant contributions to soft tissue healing [31]. In cases where bone tissue is exposed, such as alveolar osteitis, the application of A-PRF may accelerate soft tissue healing and, through support for epithelialization and immunomodulation [32], indirectly lead to reduced pain scores. According to the results of the present study, A-PRF and PBM were the two groups that presented the greatest decreases in pain scores during the follow-up period. Additionally, the A-PRF and Alveogyl groups demonstrated better results in terms of granulation tissue formation. This study's findings are consistent with the literature. In studies comparing photobiomodulation with control groups, PBM has been shown to exhibit superior analgesic effects and promote better soft tissue healing [11, 12, 22]. In the study by Al Harthi et al., which evaluated the effectiveness of Alveogyl, Alveogyl combined with PBM, and PBM alone for pain relief for alveolar osteitis, the combination of Alveogyl and PBM was found to be the most effective approach [33]. During the first three days, the pain levels in the Alveogyl + PBM group were greater than those in the Alveogyl alone group. However, at days three and four, similar results were observed between the PBM and Alveogyl + PBM groups. No significant differences in pain relief were found between the Alveogyl and PBM groups. Eshghpour et al. found that Alveogyl demonstrated greater efficacy than PBM in the initial two days following treatment, while also noting that PBM yielded acceptable VAS scores in the management of alveolar osteitis [34]. Similarly, in this study, the mean VAS scores on the second day were comparable between the PBM and Alveogyl groups. In the present study, the PBM group demonstrated superior soft tissue healing based on LHI data compared with the other groups but showed similar analgesic efficacy. Importantly, this study involved interventions rather than allowing the healing process to occur naturally in each group. Therefore, we believe that our findings contribute positively to the literature on this topic.

Sex was a significant factor influencing pain outcomes in the present study. The mean VAS score was greater among female patients than male patients, both preoperatively and on the 4th postoperative day. This result aligns with the literature indicating a higher incidence of alveolar osteitis in females [35]. Similarly, in this study, a greater proportion of female patients were observed, with females experiencing more severe pain from alveolar osteitis than males did.

Alveolar osteitis is commonly reported to increase with age, particularly affecting individuals in their third and fourth decades of life. However, the present study yielded a contrary finding in which alveolar osteitis cases were predominantly observed in individuals aged 18–27 years (33.8%). This discrepancy may be attributed to the higher prevalence of third molar extractions in young adults, as alveolar osteitis is most frequently associated with third molars [36]. In the present study, the incidence of alveolar ostieitis was similar between the 1st + 2nd mandibular molars and 3rd molars.

Interestingly, while the A-PRF group had a mean age over 40 years, the PTX group included younger patients. This variation in age distribution may have inadvertently influenced the present study's results. Future studies could benefit from the use of randomization methods to balance age distributions across groups, potentially mitigating such influences.

The methods investigated in the present study—Alveogyl, PBM, and A-PRF—have demonstrated effectiveness in treating alveolar osteitis compared with control groups in previous studies [12, 16, 37]. In this study, we compared these effective methods against each other, and the alveogil group served as a positive control group due to its inclusion in accepted treatment protocols [38]. This factor likely contributed to the absence of significant differences observed between the groups during certain periods. Patients in the Alveogyl group initially presented with higher preoperative soft tissue values than those in the other groups did, suggesting potentially better baseline healing conditions. However, PBM and A-PRF exhibited accelerated healing trajectories in terms of improvement over the follow-up period.

In the present study, patients with alveolar osteitis involving the mandibular first, second, or third molars were included. This decision was influenced by several factors, including the need for standardization, the physiological advantages of the mandible over the maxilla, gravitational benefits, and the relatively smaller wound area of the anterior and premolar teeth. Although this approach could be perceived as a limitation, efforts were undertaken to minimize local and anatomical variations, thereby enabling an objective evaluation of the effectiveness of the treatment methods. A limitation of this study is the inability to achieve an equal distribution of age groups. Given that individuals aged 40 and older may experience disadvantages in wound healing, the distribution was adjusted accordingly [39, 40]. Consequently, the mean ages across the groups were not very similar. A distribution over more blocks could have been achieved by narrowing the age ranges further. Although excluding individuals over 40 years of age might have mitigated this concern, we chose to include them to ensure better population representation. Despite the higher average age in this cohort, A-PRF demonstrated notable efficacy. While the older average age in the PTX group may have provided a potential advantage for wound healing, the clinical outcomes were not superior to those observed in the other groups.

Given that alveolar osteitis is painful, it would have been unethical to withhold analgesics. Therefore, all patients received analgesic treatment, which likely impacted the pain data. To mitigate this effect, we standardized treatment by administering the same analgesic with mild anti-inflammatory properties to all patients.

The symptomatic period for alveolar osteitis typically lasts 7 to 10 days [38, 41]. Most studies follow up for approximately two weeks, so we also assessed soft tissue healing over this duration [13]. Pain typically diminishes during the initial week; therefore, assessments concentrated on this early period. While extended follow-up could have been implemented, it was deemed preferable to forgo this option to minimize the potential for increased clinical visits and the associated risk of data loss.

Conclusion

The results of this study indicate that A-PRF and PBM are more beneficial for the treatment of alveolar osteitis than the other treatments are. In addition, we investigated the potential of PTX as an alternative treatment and concluded that it does not provide significant advantages over the other methods examined. Additionally, future clinical studies may yield different results when pentoxifylline is used in combination with existing protocols.

Abbreviations

A-PRF	Advanced Platelet Rich Fibrin
PBM	Photobiomodulation
PTX	Pentoxifylline
VAS	Visual analog scale
LHI	Landry Healing Index
sd	Standard deviation

Authors' contributions

Concept: C.N.E.; Data Collection or Processing: A.Y., C.N.E; Analysis or Interpretation: A.Y.; Literature Search: A.Y., C.N.E; Writing: C.N.E., A.Y.; Project Supervision: C.N.E. All authors reviewed the manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The study was based on the approval of the Clinical Studies Ethics Committee of the Faculty of Medicine of Akdeniz University (KAEK 21.09.2022/253–26.04.2023/155), the Turkish Medicines and Medical Devices Agency (30.05.2023/1120566). The written consent obtained from all of the participants was informed.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

[1] Department of Oral and Maxillofacial Surgery, Faculty of Dentistry, Akdeniz University, Antalya 07058, Turkey.

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